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Microstructure and mechanical properties of refractory MoNbHfZrTi high-entropy alloy

N.N. Guo, L. Wang *, L.S. Luo, X.Z. Li, Y.Q. Su *, J.J. Guo, H.Z. Fu

National Key Laboratory for Precision Hot Processing of Metals, School of Materials Science and Engineering, Harbin Institute of Technology, Harbin 150001, China

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ABSTRACT

The present paper reports the microstructure, phase stability and mechanical properties of a new refractory MoNbHfZrTi high-entropy alloy. MoNbHfZrTi alloy consists of a disordered body-centered cubic (BCC) solid solution phase in as-cast and homogenized states. Homogenization treatment and DSC analysis indicate that there is no any phase transition below 1743 K. At room temperature, the compressive yield strength for the alloy in as-cast and as-homogenized states reaches about 1719 MPa and 1575 MPa, respectively and the fracture mechanism is brittle quasi-cleavage fracture. At elevated temperatures, this alloy has compression yield strength of 825 MPa at 1073 K, 728 MPa at 1173 K, 635 MPa at 1273 K, 397 MPa at 1373 K and 187 MPa at 1473 K and some fine grains form at grain boundaries due to partial dynamic recrystallization.

1. Introduction

The demand for materials suitable for elevated temperature applications in the aircraft and aerospace industry beyond the realm of Ni-based superalloys has generated significant research interest in refractory metals and alloys [1]. Refractory metals and alloys have the characteristics of high melting point, high elevated-temperature strength, high creep resistance, good corrosion resistance, the plastic processing property and high application temperature up to 1373–3593 K, that is much higher than that of most superalloys [2]. Most of the conventional refractory metals or alloys are based on one principal element and a small amount of other elements are added to improve the properties. In the past several years, new alloy systems named high-entropy alloys have been proposed and attracted much attention [3–5]. High-entropy alloys can possess simple phases and excellent properties such as high hardness, superior resistance to temper softening, good wear resistance, resistance to oxidation and corrosion due to the four characteristics of high mixing entropy, severe lattice distortion, sluggish diffusion and cocktail effects [6–8]. Since the introduction of the high-entropy alloys, the most commonly used elements are Al, Fe, Cu, Co, Ni, Cr, Ti, V and Mn [9–12]. Since 2010, refractory high-entropy alloys on refractory elements have been proposed to explore new alloys for elevated-temperature applications [13]. For example, Senkov et al. prepared near equiatomic-concentration WTaNbMo and WTaNbMoV alloys with single disordered BCC solid solution phase exhibited clearly superior mechanical properties such as high yield strength of 405 MPa and 477 MPa at 1600 °C, compared with conventional superalloys [13,14]. Furthermore, TaNbHfZrTi alloy was prepared by substituting W and Mo for Ti and Zr for reducing the density. TaNbHfZrTi alloy exhibited significant increase of the compression strain (up to 40% compression strain at room temperature) [15]. Since then, more researchers began to explore and investigate other refractory high-entropy alloys [16–19]. A series of refractory high-entropy alloys were reported such as $NbTiVTaAl_x$, Cr–Nb–Ti–V–Zr, $NbCrMoTiAl_{0.5}$, $NbCrMoVAl_{0.5}$, $NbCrMoTiVAl_{0.5}$ and $NbCrMoTiVAl_{0.5}Si_{0.3}$ [17–25]. Most reported refractory high-entropy alloys are mainly composed of simple BCC solid solution phase. Much research indicates the design concept of refractory high-entropy alloys provides an effective method for developing the new elevated temperature alloys.

Among the reported refractory high-entropy alloy, TaNbHfZrTi alloy has a single BCC phase. The density of Ta element is 16.65 g/cm^3 and the density of Mo element is 10.28 g/cm^3. By replacing Ta in TaNbHfZrTi alloy with Mo, the density and cost of the new alloy MoNbHfZrTi are reduced. The phase, microstructure and mechanical properties at room temperature and elevated temperatures of MoNbHfZrTi alloy are investigated and discussed. This new alloy also forms a single BCC phase and shows improved compressive strength relative to TaNbHfZrTi alloy.

* Corresponding authors.
E-mail addresses: wliang1227@163.com (L. Wang), suyq@hit.enu.cn (Y.Q. Su).

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2. Experimental procedures

Alloy ingots of MoNbHfZrTi with equiatomic concentration were prepared by arc melting in a water-cooled copper crucible in high-purity argon atmosphere. All elements have the purity higher than 99.6% weight percent. All ingots were melted for five times to improve homogeneity. The ingots are about 35 mm of diameter and 15 mm of height. All samples tested and analyzed are cut from the middle part of the ingots. The as-cast samples were sealed in a vacuum quartz tube and were homogenized at 1373 K for 10 h, subsequently furnace-cooled to room temperature. The crystal structure was investigated using a D/max-rB X-ray diffractometer with Cu Kα diffraction generated at 45 kV and 40 mA with the scanning angles ranging from 20° to 100° at a scanning rate of 5°/min. The experimental compositions were analyzed on the non-etched surface by the scanning electron microscope (SEM) energy dispersive spectroscopy (EDS). To observe the microstructure, the sample was etched using the hydrofluoric acid, nitric acid and distilled water ($HF:HNO_3:H_2O$ = 1:1:8). After etching, the microstructure and chemical compositions were analyzed by SEM and EDS. The room temperature compressive properties were evaluated using an Instron 5569 testing machine in air. The elevated temperature compressive properties were evaluated using the dynamic thermal simulation testing machine (Gleeble-1500D) and the furnace chamber was evacuated to 10^{-3} torr before each test. The strain rate is $1 \times 10^{-3}\ s^{-1}$. The test specimens are in the shape of a cylinder (φ4 mm × 6 mm). The microstructure and fracture surface were investigated by the scanning electron micrograph (SEM) and the electron back-scatter diffraction (EBSD). The thermal analysis was carried out to verify the stability of the phase. The thermal analysis was carried out in SETSYS EVOLUTION 16/18 differential scanning calorimeter (DSC) at a heating rate of 10 K/min under flowing high purity argon atmosphere.

3. Results

3.1. Crystal structure

X-ray diffraction patterns of as-cast and as-homogenized states for MoNbHfZrTi alloy are shown in Fig. 1. Only one BCC phase with the lattice parameter of 338.1 pm is identified for MoNbHfZrTi alloy in the as-cast state. After homogenization at 1373 K for 10 h, MoNbHfZrTi alloy is still composed of the single BCC phase with the lattice parameter of 337.1 pm. In both as-cast and as-homogenized states, no superlattice diffraction peaks as an evidence of crystal ordering or other peaks are observed. According to the JCPDS cards (65-7192), the multiple peaks match well with the BCC ZrTiNb with the lattice parameter of 336.0 pm. So the crystal structure of as-cast MoNbHfZrTi alloy is the same as that of the BCC ZrTiNb phase.

To further confirm the phase structural stability of the disordered BCC phase in the MoNbHfZrTi alloy, DSC analysis is carried out as shown in Fig. 2. There are no any endothermic peaks or exothermic peaks on the heating curve or the cooling curve, which indicates that there is no any phase transition from 296 K to 1743 K.

3.2. Microstructure and chemical compositions

The microstructure of MoNbHfZrTi alloy in the as-cast and as-homogenized states is shown in Fig. 3. Before etching, MoNbHfZrTi alloy in the as-cast state presents no obvious feature from the SEM back-scattered images as shown in Fig. 3(a). The EDS analysis shows that the overall experimental compositions are very similar to the nominal compositions as shown in Table 1. And the chemical distribution is very homogeneous which can be verified from the EDS maps as shown in Fig. 4. After etching, the typical dendrite and interdendrite structure is observed. The interdendrites are corroded and some fine white particles are observed in the interdendrites. The white particles have the average size of 0.5–1 μm. After etching, the dendrites are enriched with Hf and the compositions of the white particles in the interdendrites are very close to that of the dendrites. The compositions of the interdendrites can be deduced from the overall composition and compositions of the dendrites and white particles. The interdendrites are enriched with Ti. After homogenization at 1373 K for 10 h, the MoNbHfZrTi alloy also presents no any obvious feature before etching as shown in Fig. 3(c). After etching, the MoNbHfZrTi alloy shows the polycrystalline structure.

3.3. Compressive properties

The compressive curves of the MoNbHfZrTi alloy at different temperatures are shown in Fig. 5. The compressive properties such as maximum strength, σ_p, yield strength, $\sigma_{0.2}$, and fracture strain, δ, are given in Table 2. The yield strength of MoNbHfZrTi alloy in as-cast state is 1719 MPa at room temperature while homogenization treatment leads to the decrease of the strength to 1575 MPa. At elevated temperatures, the alloy reaches the maximum strength at the initial stage of deformation, and strain softening occurs in the next deformation.

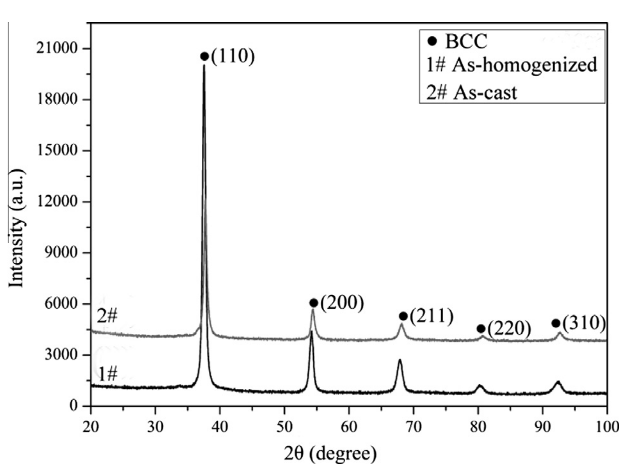

Fig. 1. X-ray diffraction patterns of the MoNbHfZrTi alloy.

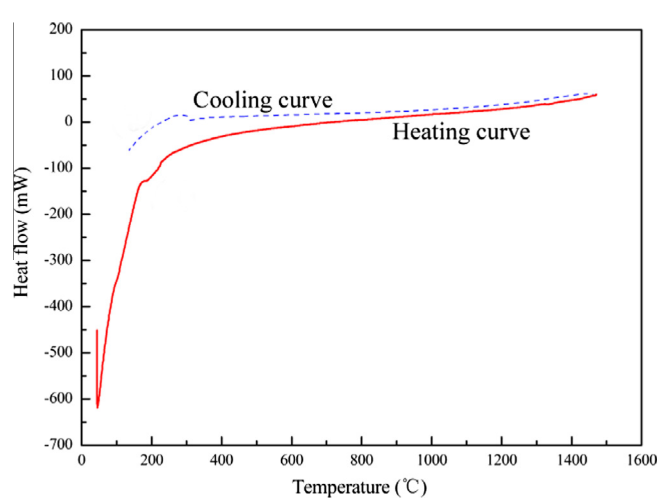

Fig. 2. DSC curves of the MoNbHfZrTi alloy.

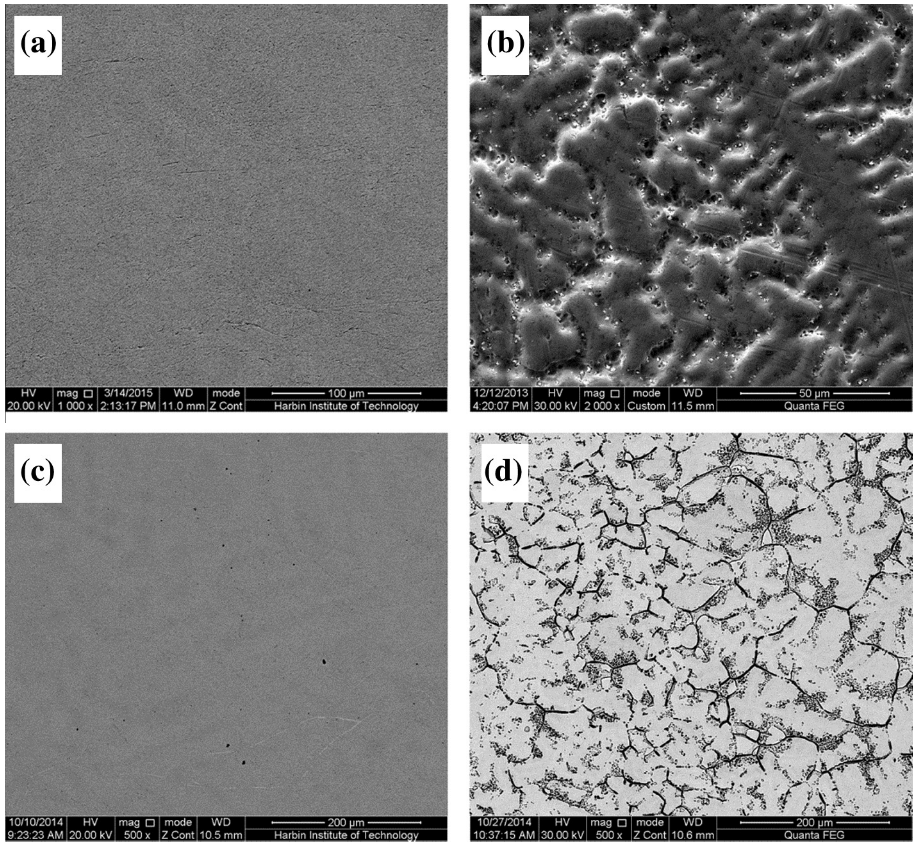

Fig. 3. (a) SEM back-scattered image of MoNbHfZrTi alloy in as-cast state before etching, (b) SEM secondary electron image of the alloy in as-cast state after etching, (c) SEM back-scattered image of the alloying as-homogenized state before etching, and (d) SEM secondary electron image of the alloy in as-homogenized state after etching.

Table 1
Chemical compositions (at.%) of MoNbHfZrTi alloy in as-cast state.

	Mo	Nb	Hf	Zr	Ti
Nominal	20	20	20	20	20
Experimental	19.58	20.51	20.42	20.10	19.39
Dendrites	18.39	20.16	26.01	19.86	15.58
White particles	19.23	19.20	26.24	20.82	15.50

Fig. 6 shows the microstructure of the MoNbHfZrTi alloy after compressive deformation at different temperatures. At room temperature, the fracture surface exhibits apparently cleavage steps, river patterns and tongue patterns. After compression at 1073 K, MoNbHfZrTi alloy exhibits the polycrystalline structure. And the grains are elongated in the directions of plastic flow, which are inclined by 90° to the compression direction as shown in Fig. 6(c). After compression at 1173 K, the alloy also exhibits the polycrystalline structure and some dark fine particles or grains form at grain boundaries as shown in Fig. 6(d). With the increase of the compressive temperature, the polycrystalline grains become more elongated and grow larger. Fig. 7 shows the XRD pattern of this alloy after deformation at 1273 K. MoNbHfZrTi alloy still remains the single BCC phase with the lattice parameter of 337.5 pm. It can be deduced that the dark fine particles or grains at grain boundaries are not new phase and they may be fine dynamic recrystallized grains.

4. Discussion

4.1. Phase formation

The strictly definition of high-entropy alloys is those alloys with a disordered solid solution phase containing multiprincipal elements in an equiatomic or a near-equiatomic concentration, whose configurational entropy is tremendously high [6]. MoNbHfZrTi alloy with the single BCC disordered solid solution phase meets well with the definition of the high-entropy alloy. Thus this alloy is one of high-entropy alloys deserved to study. While according to the much research on the high-entropy alloys, the high configurational entropy is not the only factor for the formation of the solid solution phases [26,27]. Both the physical and chemical characteristics of the components and their interactions such as the mixing enthalpy are also the dominant factors for the formation phases. For MoNbHfZrTi alloy, all five elements in this alloy have a BCC crystal structure and mutual solubility except Mo–Hf and Mo–Zr binary systems at elevated temperatures. The binary atomic radius difference of these 5 elements is less than 15% and they have similar valence and electronegativity except for Mo. (The characteristics of five components are listed in Tables 3 and 4.) According to the Hume-Rothery rules, the MoNbHfZrTi alloy does not completely meet the constraints of forming a disordered solid solution phase. And it has also been found that the constraints set by Hume-Rothery rules seem to be probably relaxed for some high-entropy alloys. Therefore, researchers have proposed some other parameters to predict the phase formation.

Zhang et al. have defined two parameters, δ_r and Ω, which are the atomic size difference and the combined effects of the mixing entropy (ΔS_{mix}), mixing enthalpy (ΔH_{mix}) and melting temperature (T_m), to predict the phase formation for multiprincipal-component alloys [26,28,29]. $\Omega \geqslant 1.1$ and $\delta_r \leqslant 6.6\%$ are required to form solid-solution phases. For MoNbHfZrTi alloy, the values of δ_r and Ω are 5.95% and 20.27%, respectively, which meets the criterion of the formation of the solid solution phase. The electronegativity

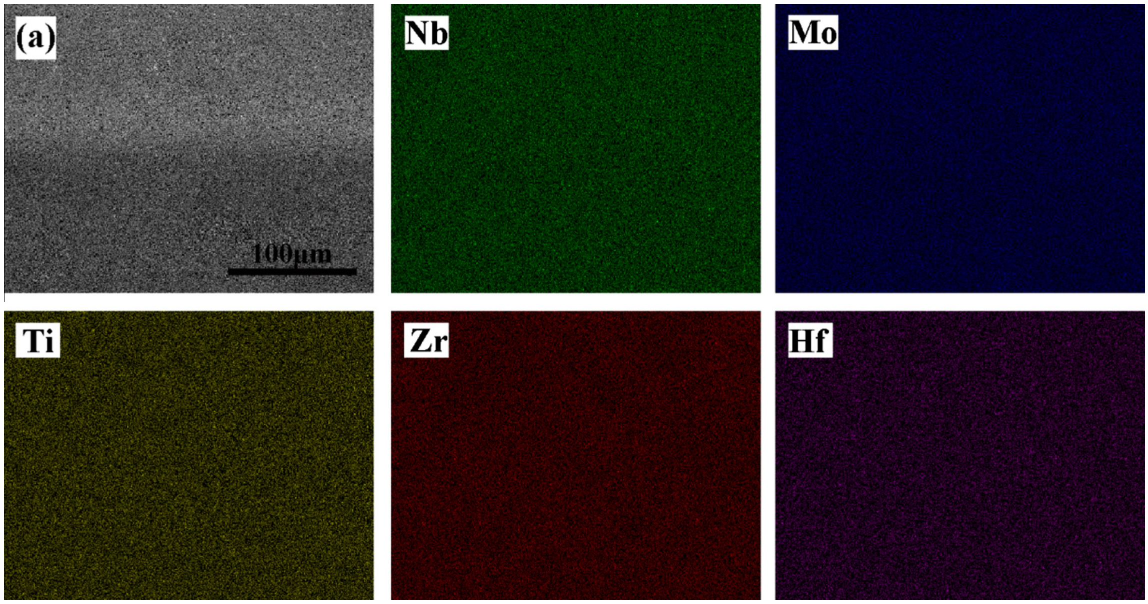

Fig. 4. EDS maps of components for MoNbHfZrTi alloy in the as-cast state.

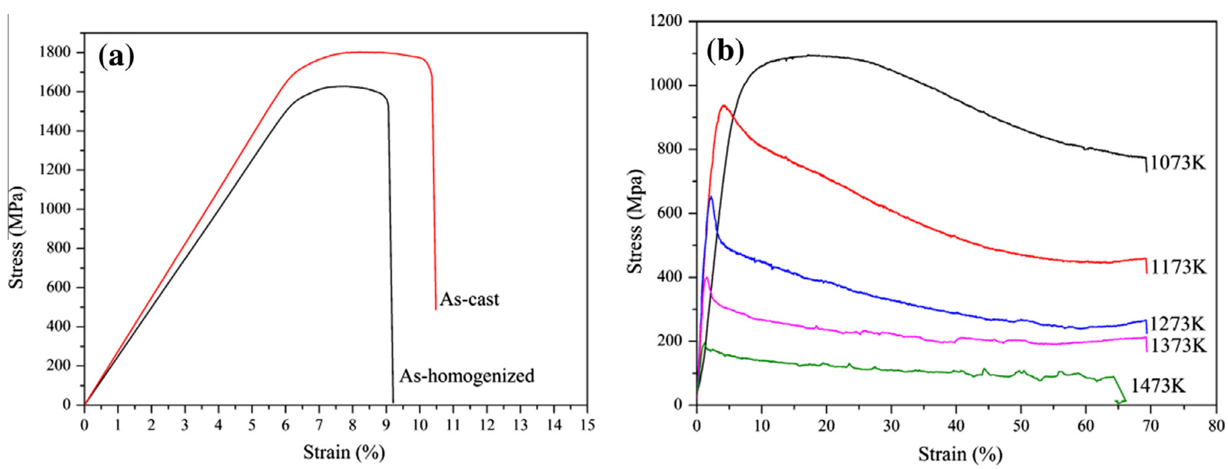

Fig. 5. Compressive curves of MoNbHfZrTi alloy: (a) at room temperature, and (b) at different elevated temperatures.

Table 2
Compressive properties of maximum strength, σ_p, yield strength, $\sigma_{0.2}$, and fracture strain, δ at different temperatures.

T (K)	296-C	296-H	1073	1173	1273	1373	1473
σ_p (Mpa)	1803	1640	1095	938	654	399	194
$\sigma_{0.2}$ (Mpa)	1719	1575	825	728	635	397	187
δ (%)	10.12	9.08	>60	>60	>60	>60	>60

difference, δ_χ, has also been proposed to predict the formation of the phase for multi-principle component alloys. Although there is no specific range for the formation of the solid solution phase, a lower value of δ_χ is favorable for the formation of the solid solution phase [30]. The δ_χ for MoNbHfZrTi alloy is 19.96%. Compared with the values of the δ_χ for other high-entropy alloys, the value of δ_χ for MoNbHfZrTi alloy is not very low. The δ_χ is not an effective parameter for predicting the formation of the solid solution phase. Guo et al. have proposed valence electron concentration (*VEC*) to determine the phase stability for FCC or BCC solid solutions. FCC solid solution phases will be more stable at higher *VEC* ($\geqslant$8), a low *VEC* (<6.87) will stabilize the BCC solid solution phases [31]. The value of *VEC* for MoNbHfZrTi alloy is 4.6, which is lower than the boundary value. The parameters of MoNbHfZrTi alloy basically meet the existing criteria. In the next study, these existing can be used for optimizing MoNbHfZrTi alloy.

4.2. Phase stability

MoNbHfZrTi alloy is composed of a single BCC phase in as-cast and as-homogenized states and the DSC result also verifies that there is mo any phase transition below 1743 K. It can be concluded that the single BCC phase has high structural stability. The decrease of the lattice parameter may be attributed to the following reason: As-cast alloy solidified in the water-cooled copper crucible in which the cooling rate is high. After solidification, the internal stresses generate in as-cast alloy, leading to the increase of the lattice distortion. Homogenization treatment may release the internal stress partly and decreases the lattice distortion.

Using the rule of mixtures, a theoretical lattice parameter (a_{mix}), of an alloy with the disordered BCC phase can be estimated:

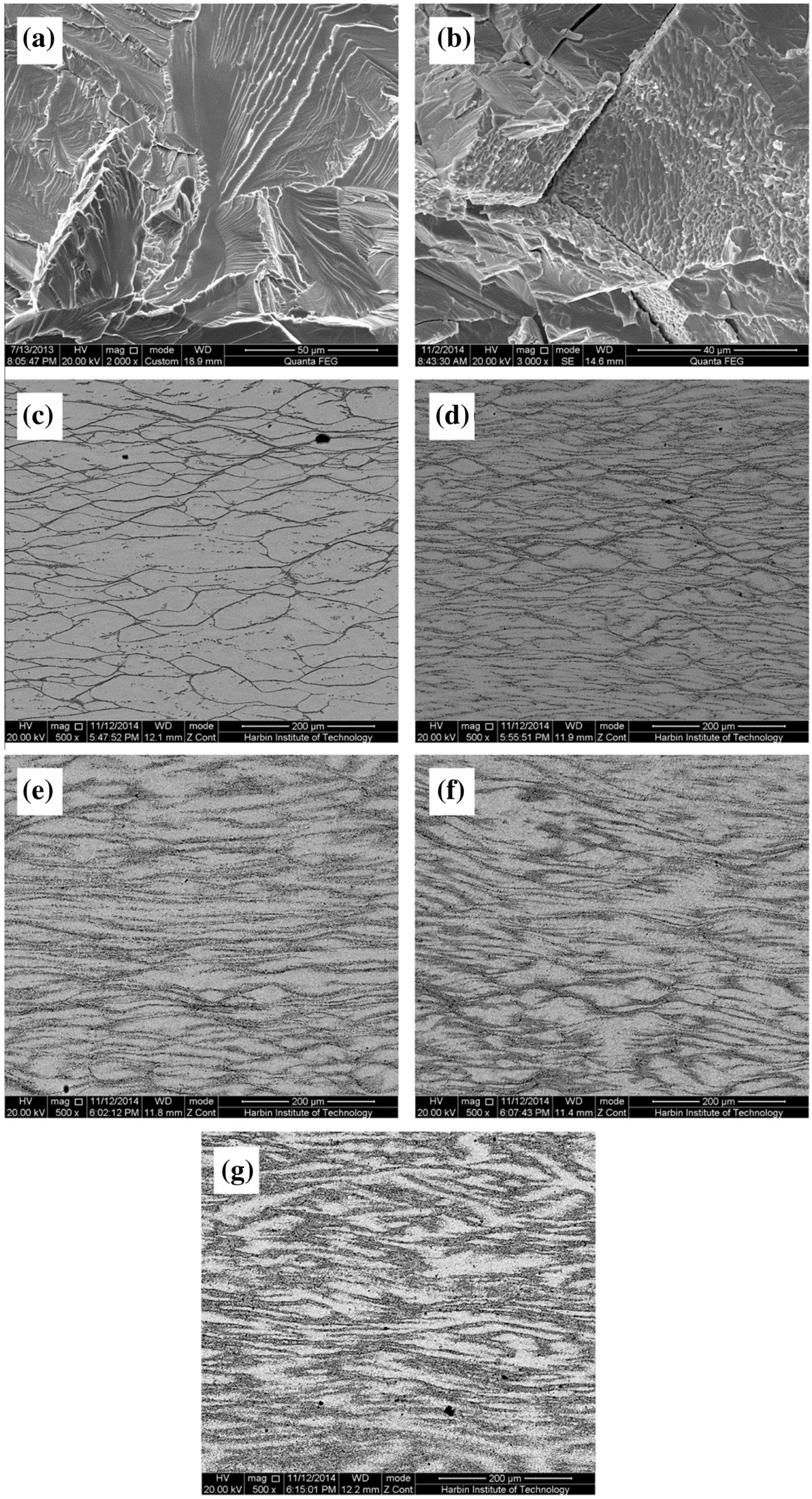

Fig. 6. Images of the microstructure of MoNbHfZrTi alloy after compressive deformation: (a) in as-cast state at room temperature, (b) in as-homogenized state at room temperature, (c) at 1073 K, (d) at 1173 K, (e) at 1273 K, (f) at 1373 K, and (g) at 1473 K.

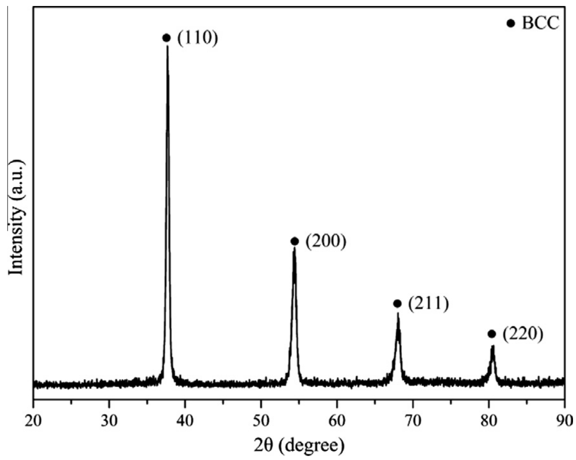

Fig. 7. X-ray diffraction pattern of MoNbHfZrTi alloy after deformation at 1273 K.

$$a_{mix} = \sum_{i=1}^{n} c_i a_i$$

where a_i is the lattice constant of element *i*. Three elements, Ti, Zr, and Hf have a hexagonal close packed (HCP) at room temperature. Their BCC lattice parameters listed in Table 3 at room temperature can be calculated according to their BCC lattice parameters at elevated temperature and their thermal expansion coefficients [15]. The value of the theoretical lattice parameter for this alloy is 337.3 pm. The experimental lattice parameters, especially for the alloy in as-homogenized state, are closed to the theoretical lattice, which indicates this alloy follows the rule of mixtures. This analysis may indicate that the BCC phase is a fully disordered solid solution phase.

4.3. Compressive properties

The compressive yield strength is high for MoNbHfZrTi alloy in as-cast state at room temperature and elevated temperatures. For the high entropy alloy with single solid solution phase, the high strength is mainly owing to the solution strengthening. Five elements have equiatomic concentration and each atom can be viewed as the solute atoms or solvent atoms. All atoms with different sizes and properties will interact with each other and elastically distort the crystal lattice, which induces the formation of a local elastic stress field, to hinder dislocation movements and cause the increase of strength. After homogenization, this alloy still remains the same BCC phase while the compressive yield strength decreases slightly. Homogenization treatment releases the internal stress generated from the high cooling rate and decreases the lattice distortion, that leading to the decrease of strength. And the solution strengthening is not the only strengthening mechanism. After homogenization treatment, the microstructure of MoNbHfZrTi alloy becomes coarse as shown in Fig. 3(b) and (d).

Table 4
The values of mixing enthalpy (ΔH_{mix}, KJ/mol) calculated by Miedema's model for binary atomic pairs.

Element	Mo	Nb	Hf	Zr	Ti	Ta
Mo	0	−6	−4	−6	−4	−5
Nb		0	4	4	2	0
Hf			0	0	0	3
Zr				0	0	3
Ti					0	1
Ta						0

The fine–grain strengthening is also one contributor of the higher strength. From the fracture morphology as shown in Fig. 6(a) and (b), brittle quasi-cleavage fracture is evident.

Fig. 8 shows the EBSD map of MoNbHfZrTi alloy after deformation at 1073 K and 1473 K. After deformation at 1073 K, some fine dynamic recrystallized grains are observed. The result confirms the previous conclusion that the dark particles or grains are dynamic recrystallized grains. After deformation at 1473 K, polycrystalline structure become coarser and more dynamic recrystallized grains form. These dynamic recrystallized grains are mainly responsible for the decrease of strength. The volume fraction of the dynamic recrystallized grains increases from 10.2% at 1073 K to 27.5% at 1473 K. And the EBSD data is used to determine the average misorientation distribution as shown in Fig. 9. There are a high fraction in the low-angle boundary regime (<5°).

Compared with the reported TaNbHfZrTi alloy [32], the alloy remains a single BCC phase after replacement Ta with Mo while the strength increases whether at room temperature or at elevated temperatures though the plasticity decreases at room temperature. It may be attributed to the following reasons: (1) the binary atomic radius difference of Mo with Nb, Ti, Zr, Hf elements is larger than that Ta with these four elements. Mo element will cause more serious lattice distortion and presents more prominent solution strengthening effect. (2) The mixing enthalpy of Mo with Nb, Ti, Zr, Hf elements is more negative than that of Ta with other four elements. That means the binding energy of Mo with Nb, Ti, Zr, Hf elements is higher.

5. Conclusions

In this paper, a new refractory high-entropy alloy, MoNbHfZrTi, was explored and prepared. The microstructure, phase stability and mechanical properties were studied. Based on the obtained results and analysis, four conclusions can be drawn as follows.

(1) MoNbHfZrTi alloy is composed of a single disordered BCC phase in as-cast and as-homogenized states. The observed phase in this alloy can be predicted using the existing criteria.
(2) The homogenization and DSC analysis confirm that there is no any phase transition below 1743 K for MoNbHfZrTi alloy with the single BCC phase. The structural stability makes this alloy be a promising candidate for elevated-temperature applications.

Table 3
Characteristics of Nb, Mo, Ti, Zr, Hf (A2: BCC, A3: HCP, the lattice parameter of BCC crystal structure at room temperature [15]).

Element		Mo	Nb	Hf	Zr	Ti	Ta
Crystal structure	High temperature	A2	A2	A2	A2	A2	A2
	Low temperature			A3	A3	A3	
Atomic radius, r, (pm)		136.3	142.9	157.8	160.3	146.2	1.430
Lattice parameter, a, (pm)		314.7	330.1	355.9	358.2	327.6	330.3
Pauling electronegativity, χ		2.16	1.60	1.30	1.33	1.54	1.50
Valence electron concentration, V		5	5	4	4	4	5
Melting temperature, T_m, (K)		2896	2750	2506	2128	1941	3268

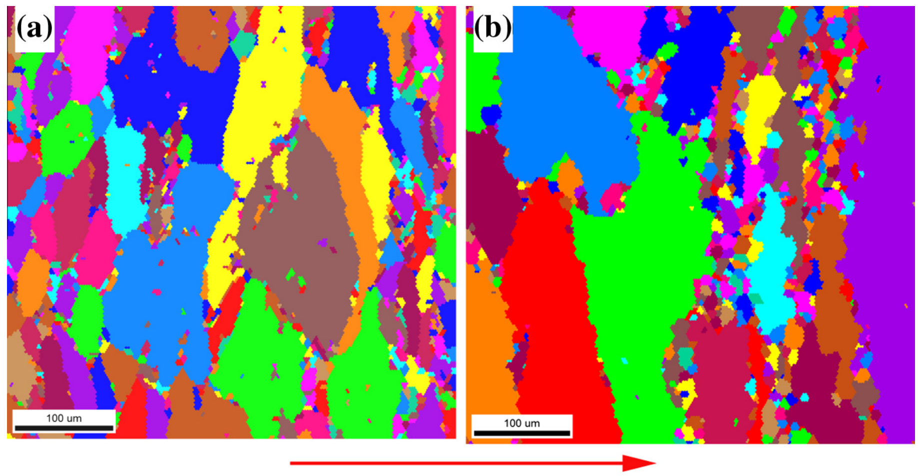

Fig. 8. EBSD maps of MoNbHfZrTi alloy after deformation: (a) at 1073 K, and (b) 1473 K, the red arrow shows the compressive direction. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

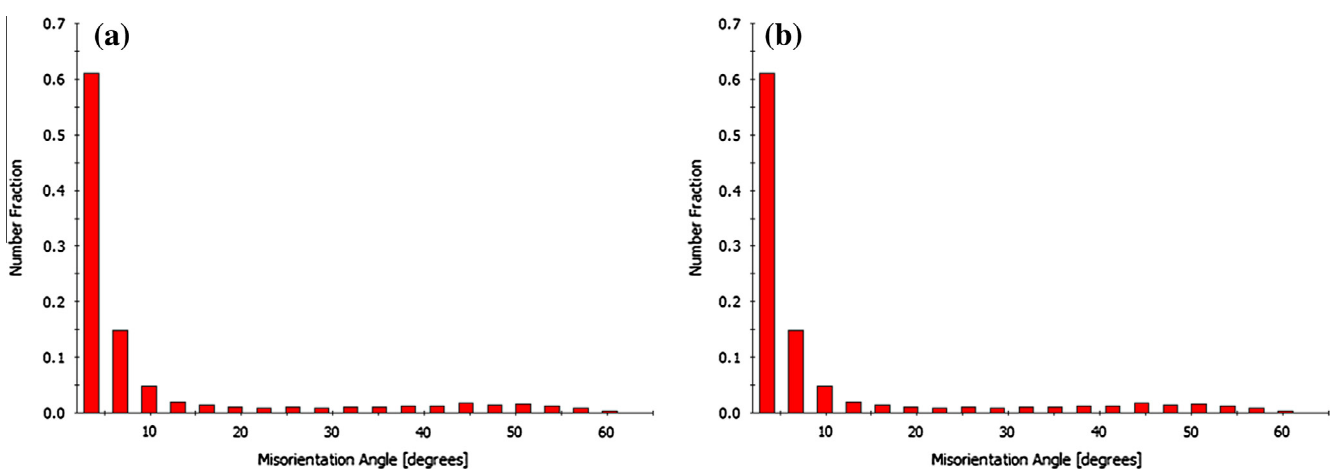

Fig. 9. Misorientation angle distributions of as-cast alloys after deformation: (a) at 1073 K, and (b) at 1473 K.

(3) At room temperature, the alloy has high compressive yield strength of 1719 Mpa and 1575 Mpa in as-cast and as-homogenized states. The fracture mechanism is quasi-cleavage fracture from the features of cleavage steps, river patterns and tongue patterns.
(4) At elevated temperatures, this alloy still has high compressive yield strength (825 MPa at 1073 K, 728 MPa at 1173 K, 635 MPa at 1273 K, 397 MPa at 1373 K and 187 MPa at 1473 K) and shows a drop of flow stress after yielding. Some fine dynamic recrystallized grains form at the grain boundaries which is responsible for the strain softening.

Acknowledgements

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